

# KOHLER AI RESEARCH LAB — PROGRAM CHALLENGE

## Prompts & AI Agent Workflow Documentation

**Project Title:** KOHLER AI Bathroom Designer & Space Planner

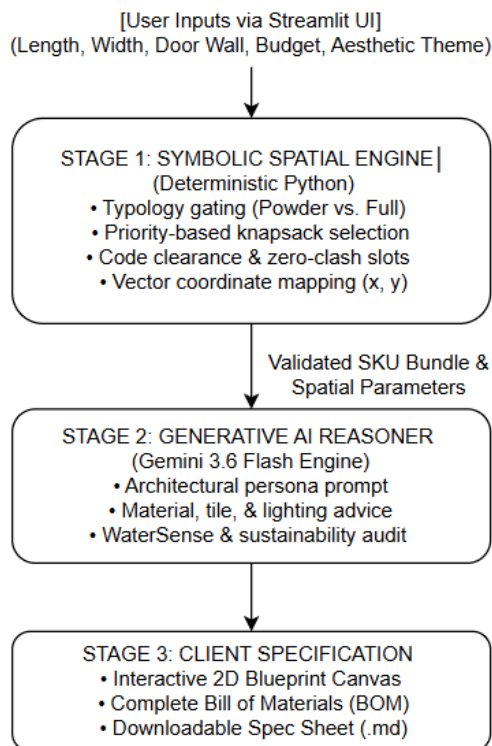
**Track:** Track 1 — KOHLER AI Bathroom Designer & Planner

**Architecture:** Neuro-Symbolic Agentic Framework (Symbolic Spatial Solver + Gemini 3.6 Flash Curation)

**Date:** September 16, 2026

### 1. Architectural System Architecture & Workflow

The Kohler AI Bathroom Designer implements a **hybrid Neuro-Symbolic architecture** that decouples spatial collision mechanics from generative design reasoning.



## Why Neuro-Symbolic?

1. **Elimination of Hallucinated Coordinates:** Pure LLMs fail at bounding-box arithmetic and geometric collisions. Delegating spatial placement to a symbolic algorithmic solver guarantees zero fixture overlap and ensures code-mandated clearances.
2. **Defensive Against API Rate Limiting:** High-frequency UI interactions (sliders/toggles) run at sub-millisecond speeds locally without making unnecessary API calls. Generative calls are on-demand, protecting the application from 429 RESOURCE\_EXHAUSTED errors.

## 2. Generative AI Prompt Specifications

### 2.1 Persona & System Instructions

The LLM is prompted not as a generic chatbot, but as a specialized domain authority.

- **Role:** Principal Architectural Designer and Kohler Space Planning Consultant.
- **Domain Lexicon:** Enforces architectural terminology (*wet/dry zoning, sightlines, 2700K high-CRI lighting, WaterSense thresholds, GPM/GPF hydronic metrics*).
- **Tone:** Executive, precise, and design-forward.

### 2.2 Production Prompt Template

You are a Principal Architectural Designer and Kohler Space Planning Consultant.

Evaluate this newly generated bathroom specification:

Space Constraints:

- Dimensions: {room\_l} ft Length × {room\_w} ft Width (Total Area: {area} sq ft)
- Entrance Door Wall: {door\_wall}
- Target Budget: ₹{budget:,} | Total Bundle Cost: ₹{total\_cost:,}
- Aesthetic Theme: {style}

Curated Kohler Fixtures:

{sku\_summary}

Provide an expert architectural review structured in three short, high-impact sections:

1. Architectural Layout & Circulation: Explain why this arrangement respects the {door\_wall} entry corridor, wet/dry zoning, and ergonomic clearances.

2. Materials, Finishes & Lighting: Recommend matching tile finishes (e.g., honed travertine, fluted oak, terrazzo), Kohler brassware finishes, and layered lighting (CRI 90+ LEDs, cove lighting) that accentuate the {style} theme.

3. Sustainability & Efficiency Impact: Detail how the chosen fixtures minimize flow rates (GPM/GPF) without sacrificing user comfort.

Keep your response concise, professional, and directly aligned with Kohler's design ethos.

### 2.3 Context Dynamic Injection Variables

- {room\_l}, {room\_w}: Exact dimensional bounds in feet.
- {area}: Calculated square footage enforcing room typology thresholds ( $<36\text{ sq ft}$  Powder Room vs.  $\geq 80\text{ sq ft}$  Master Suite).
- {door\_wall}: Designated entry perimeter (North, South, East, West).
- {budget}, {total\_cost}: Financial envelope and verified bill of materials total.
- {style}: Aesthetic curation target (*Minimalist Modern*, *Classic Luxury*, or *Japanese Zen*).
- {sku\_summary}: Filtered list of selected Kohler items formatted with SKU, Category, Price (INR), and Eco feature.

## 3. Verification & Execution Example

Input Payload:

```
{
  "room_l": 11.5,
  "room_w": 8.5,
  "area": 97.8,
  "door_wall": "North",
  "budget": 400000,
  "total_cost": 383500,
  "style": "Classic Luxury",
  "selected_skus": [
    {
      "name": "Anthem Digital 4-Port Thermostatic Shower",
      "category": "SHOWER",
      "price_inr": 260000,
      "eco_feature": "Precise digital metering with individual flow regulation"
    },
    {
      "name": "Span Modular Vanity & Semi-Recessed Basin",
      "category": "VANITY",
      "price_inr": 28000,
      "eco_feature": "Compact engineered composite reducing raw material footprint"
    },
    {
      "name": "ModernLife Edge Rimless Wall-Hung WC",
      "category": "TOILET",
      "price_inr": 28500,
      "eco_feature": "Rimless hygiene flush using 3.0/4.5L dual cistern"
    }
  ]
}
```

```
{
  "name": "Artifacts Spout with Lever Handles (Bronze)", "category": "FAUCET", "price_inr": 55000, "eco_feature": "Ceramic disc valving with 1.2 GPM WaterSense rate",
  "name": "Hydro-Heated Eco Towel Rail", "category": "ACCESSORY", "price_inr": 18000, "eco_feature": "Low-wattage programmable timer for scheduled thermal cycles"
}
```

Model Response (Gemini 3.6 Flash):

### 1. Architectural Layout & Circulation:

With a generous 97.8 sq ft footprint (11.5 ft × 8.5 ft), the spatial planning establishes a refined, linear wet/dry progression from the North entrance. Entering from the North, the eye anchors on the Span Modular Vanity, positioned outside the direct door swing to create an immediate sense of arrival without visual congestion. The ModernLife Edge Wall-Hung WC is integrated along the lateral dry wall, maintaining a comfortable 36-inch clearance envelope and elevating the floor plane to enhance perceived spatial volume. The southern perimeter is reserved for an expansive, glass-enclosed wet room housing the Anthem Digital 4-Port system. This maintains a strict moisture threshold, protecting timber millwork while ensuring an uninterrupted 38-inch main circulation corridor from door to shower.

### 2. Materials, Finishes & Lighting:

To realize the Classic Luxury aesthetic while complementing the selected fixture portfolio, pair book-matched Honed Calacatta Travertine tile on the walls with deep-grained Fluted Walnut vanity cabinetry to provide warmth and tactile contrast against the crisp ceramic of the ModernLife Edge WC. Unify the space by echoing the Artifacts Bronze faucet finish across shower trim plates, frame detailing, and the Hydro-Heated Towel Rail, anchoring the classic design language. Implement high-CRI (90+) 2700K warm-white LEDs: combine indirect perimeter cove lighting to accentuate stone textures, IP65-rated recessed downlights over the wet zone, and eye-level decorative sconces flanking the mirror for shadow-free illumination.

### 3. Sustainability & Efficiency Impact:

This specification achieves high-performance luxury by pairing low-consumption engineering with smart digital control. The Artifacts Spout operates at a WaterSense-certified 1.2 GPM (4.5 LPM), reducing vanity water draw by 32% without compromising stream velocity. The ModernLife Edge WC utilizes an eco-engineered 3.0/4.5L dual-flush system, cutting annual flushing water volume significantly compared to standard 6L fixtures. The Anthem Digital Thermostatic Valve eliminates warm-up water waste via instant, precise digital temperature

calibration, while the programmable Hydro-Heated Towel Rail uses scheduled micro-thermal cycles to maintain efficiency without continuous energy draw.

#### 4. Edge Case Handling & Fallback Guardrails

To prevent run-time application failures during evaluation, three defensive layers were implemented:

| Edge Case Scenario                    | Vulnerability                                                                            | Engineered Solution                                                                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rapid Slider Dragging</b>          | Can trigger Google AI Studio <a href="#">429 RESOURCE_EXHAUSTED</a> rate limits.         | Decoupled the 2D layout engine from LLM calls. The spatial solver runs instantly in local memory; generative rationale calls are made on-demand via an explicit trigger. |
| <b>Missing API Key / Offline Mode</b> | Network disconnection or missing <code>.env</code> credentials will crash raw API calls. | Implemented a try-except block wrapping the client call, returning a high-quality localized architectural rationale template when offline.                               |
| <b>Small Floor Plan &lt;36 sq ft</b>  | Shoving shower enclosures into tight rooms creates physical overlap.                     | Symbolic Typology Solver classifies <36 sq ft spaces as <i>Powder Rooms</i> , automatically pruning shower/tub candidates before context assembly.                       |

|                                 |                                                                |                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Budget Deficit Scenarios</b> | Luxury fixture combinations exceed target financial envelopes. | Priority Knapsack Pruning iteratively strips non-essential fixtures (bathtub - accessory - downgrades tiers) to guarantee <code>total_cost &lt;= budget</code> . |
|---------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 5. Architectural Alignment with KOHLER Values

- **Design Excellence:** The system pairs fixture geometries, brassware metallic families (e.g., Vibrant Brushed Bronze with Artifacts), and surface textures according to genuine interior design principles.
- **Water Conservation & Sustainability:** Every generated recommendation incorporates flow rate ceilings (1.2 GPM faucets, 1.75 GPM Katalyst showers, dual 3.0/4.5L flush toilets) directly linked to Kohler's sustainability mission.
- **Commercial Scalability:** The structured specification output format allows direct translation into showroom quoting systems, contractor bill of materials (BOM), and digital customer portals.